



Predicting Video Engagement and Virality: **TikTok**



Research Question

What makes a TikTok video go viral? Specifically, what factors drive video engagement, and can we build a model to predict a video's potential reach?



Data Overview



Dataset: 12,344 TikTok videos

Outcome Variables: View Count, Like Count, Share Count, Comment Count

Predictors across five categories:

- Creator metrics
- Visual quality
- Facial expressions
- Transcript emotions
- Topic flags



We take a progressive modeling approach: starting with **OLS** to establish a baseline and compare all four outcomes, applying **Ridge/Lasso** to test predictor stability and narrow our outcome focus, and finally deploying **Random Forest** to capture non-linear patterns and identify key drivers of video views.



Part 1. OLS Baseline

GOAL

Use OLS as an interpretable **baseline** to compare four engagement outcomes and identify initial key drivers of TikTok engagement. Use the OLS results to **guide the next modeling step**.

PROCESS

1. **Log Transformation:** engagement counts were highly right-skewed, so we used log-transformed outcomes to reduce the influence of viral outliers.
2. **Data Cleaning:** removed *Brightness* and *Sharpness* due to high missingness; kept *image_quality* as the main visual-quality proxy.

Methodology

Ran four separate models for: Views, Likes, Shares, Comments; compare R^2 across outcomes to understand which engagement type is best explained by the OLS baseline.

Model 1. OLS Key Finding 1



Predictor: Follower Count	Views	Likes	Shares	Comments
Coefficient	0.4296	0.1513	0.1523	0.2072
Significance	< 2e-16, ***	< 2e-16, ***	< 2e-16, ***	< 2e-16, ***

Creator scale is the most consistent driver

- **Evidence:** follower count is positive and highly significant across views, likes, shares, and comments.
- **Conclusion:** larger creators tend to generate stronger engagement across all outcomes.



Model 1. OLS Key Finding 2



Predictor: Verified.StatusTrue	Views	Likes	Shares	Comments
Coefficient	0.1794	0.2560	0.3568	-0.1427
Significance	0.060014	0.000519, ***	5.85e-08, ***	0.035526, *

Verification supports amplification more than conversation

- **Evidence:** Verified status is positively associated with likes and shares, marginally impacted views, but negatively associated with comments.
- **Conclusion:** Verified creators may help content gain approval and amplification, but they do not spark impression or discussion.



Model 1. OLS Key Finding 3



Predictor	Views	Likes	Shares	Comments
Political Topic	– **	– *	n.s.	+ *
COVID Topic	n.s.	n.s.	n.s.	+ ***
Transcript Anger	+ .	+ ***	+ *	+ ***
Transcript Surprise	+ .	+ ***	+ ***	+ ***

. $p < .10$, * $p < .05$, ** $p < .01$, *** $p < .001$; n.s. = not significant

Comments are more topic- and emotion-driven

- **Evidence:** Topic- and emotion-related predictors are positively associated with Comments, but topic related content are negatively associated with Views and Likes.
- **Conclusion:** Topic-related content drives comments but comes at the cost of views and likes. Brands chasing reach should be cautious about leaning into controversial topics.



Model 1. OLS Model Fit + Next Step



Engagement Outcome	R ²	Interpretation
Views	0.3999	Best explained by current model
Likes	0.3600	Also relatively well explained
Comments	0.2826	Less explained
Shares	0.2041	Hardest to explain with current variables

Among the four engagement outcomes, views achieved the highest explanatory power ($R^2 = 0.40$) and most directly measures content reach, which is our core research objective. For the next step, we decided to carry all four outcomes into regularization models to test predictor stability.



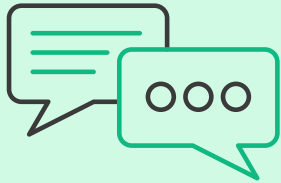
Part 2: Testing Predictive Stability



To answer what drives TikTok engagement, we tested the same creator, content, emotional, and topic predictors across four engagement outcomes: views, likes, shares, and comments.



Method	Purpose
Ridge	Keeps all predictors but shrinks coefficients to reduce overfitting → test whether the full predictor set remains stable
Lasso	Shrinks weaker predictors to zero → identify which variables are most consistently useful for predicting engagement



Part 2: Ridge & Lasso Results



Outcome	Ridge λ .min	Lasso λ .min	Lasso Variables Retained	Interpretation
Views	0.177	0.0097	18	Most simplified by lasso
Likes	0.127	0.0063	22	Most predictors retained
Shares	0.074	0.0025	23	Weak prediction, but broad predictors retained
Comments	0.095	0.0017	24	Most complex; almost all predictors retained



Across all four outcomes, the selected ridge λ values were relatively small, suggesting the model did not require extremely strong shrinkage. This means the full predictor framework was generally stable. **Lasso showed that views could be predicted with a more compact set of variables than likes, shares, or comments.**



Part 2: Key Engagement Drivers From Lasso



*We focus on views because it had the strongest model fit and retained the fewest predictors.

Driver Category	Lasso-Selected Variables for Views & Likes	Interpretation
Creator Scale	log_follower, log_likes_sum, Verified.Status	Larger and more established creators are more likely to generate views
Transcript Emotion	transcript_Anger, transcript_Joy, transcript_Surprise	Emotional cues in the transcript help predict engagement, especially emotionally charged or attention-grabbing content
Topic Signals	politic_related, gmos_related, climate_related, covid_related	Topic category shapes engagement, but the direction differs by outcome and should be interpreted cautiously
Visual / Facial Features	Limited selection	Effects are smaller and less central than creator and transcript variables

Part 2: Key Findings From Lasso



Lasso shows that views can be explained with fewer variables

The view model retained the fewest predictors, suggesting a more focused driver structure.

Creator scale is the most consistent driver

Follower count, total likes, and verified status were retained.

Transcript emotions matter for engagement

Anger, joy, and surprise were repeatedly selected by lasso.

Visual/facial features are less consistent

Smile, face detection, and image quality were not stable predictors for views.



Part 3: Random Forest Overview

Goal

- Dv: Video View
- Identify key drivers of TikTok video **views** and test whether a non-linear model outperforms OLS.

Why

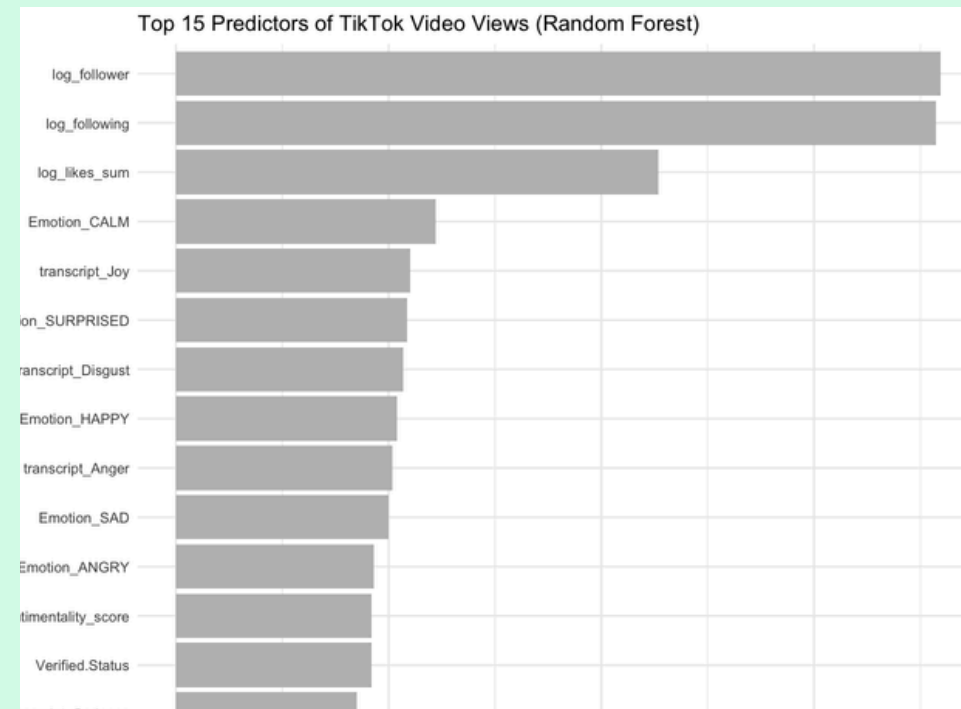
- TikTok engagement likely involves non-linear patterns and variable interactions that OLS cannot capture.
- Random Forest relaxes the linearity assumption to better reflect real-world complexity.

Methodology

- 80/20 train-test split
- 500 trees
- $mtry = 8$
- R^2 : Model performance evaluated via out-of-sample
- **%IncMSE**: to measure the Variable importance

Part 3: Random Forest -- Outputs

Variable	%IncMSE
log_follower	71.9
log_following	71.5
log_likes_sum	45.3
Emotion_CALM	24.6
transcript_Joy	22.1
transcript_Disgust	21.4
Emotion_SURPRIS	21.1
Emotion_HAPPY	20.8
transcript_Anger	20.3
Verified.Status	18.4
Emotion_ANGRY	18.6
face_detected	16.8
image_quality	16
transcript_Sadnes	16.9
transcript_Surpris	16.7



Key Drivers for Video Views:

- 1. Creator influence:*** the strongest predictor
- 2. Emotional driver:*** Calm
- 3. Topic tags:*** show near-zero predictive power

OLS R^2 : 0.3967 → Random Forest R^2 : 0.5234



Choosing the right creator matters more than crafting the perfect content, but when it comes to content, videos that convey calm or surprise emotions maximize view potential.



Part 4: Business Implications & Recommendations



Prioritize partnering with established creators

Creator scale is the strongest and most consistent predictor of reach. Brands should allocate influencer budgets toward larger, verified creators when broad viewership is the campaign goal.



Invest in calm, emotionally expressive content

Besides the creator scale, emotional tone is the strongest content predictor of views, with calm facial expressions ranking highest. Brands should prioritize composed, steady delivery over high-energy or visually polished content.



Avoid relying on topic tags to boost reach

Topical categorization has little influence on how many people see a video. Brands should focus their budget and strategy on emotional tone and creator fit rather than chasing trending or niche topics.

Part 5: Limitations & Future Research



Correlation, not causation

Observational data cannot confirm that content features directly cause higher views.



Run A/B experiments on content variables to establish causal effects.



No time dimension

Engagement metrics are a single snapshot. We cannot track how views and likes build over a video's lifetime.



Collect longitudinal panel data to model engagement growth over time.



Algorithm is a black box

TikTok's recommendation system is unobserved. View counts may reflect algorithmic push, not content quality alone.



Incorporate API-level data, such as traffic source, to separate organic from algorithmic reach.





Thank
You

